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Secure Life General Hospital Performance Report

An Analysis for the hospital management to monitor performance and make informed decisions

Introduction

Secure Life General Hospital records every patient visit, bill and insurance claim, but management has no report showing the hospital's performance. This project turns two years of raw visit records, January 2024 to December 2025 (by admission date), into an interactive Excel dashboard. This dashboard tracks patients' admissions, departmental workload, billing, insurance coverage and payment status. This will enable the management to spot problems early and make better decisions.

Tools used: Microsoft Excel (Formulas, PivotTables, Pivot Charts, slicers and dashboard design)

About the Dataset

The dataset contains 1,000 patient visits and 15 columns. Each row is one visit, not necessarily one unique patient. Missing values do not exist in this dataset. The 1,000 visits carry 684 distinct patient names, which suggests some patients visited more than once, but there is no unique patient identifier to confirm this.

Data quality checks:

- No duplicate rows or duplicate Patient IDs.
- No discharge date falls before its admission date, and every stay is at least one day.
- Ages range from 1 to 85, so no infants under 1 year are recorded. All Pediatrics visits are children (1–17), and all children were treated in Pediatrics.
- Length of Stay, Insurance Amount and Age Group were recalculated independently and matched the columns used in the analysis.
- Visits are assigned to a year by admission date. Admissions run from 1 January 2024 to 28 December 2025, and the latest discharge is 3 January 2026.

Here are terms used in this dataset and their meanings:

Terms	**Meaning**
Patient ID	Unique visit ID for every patient
Patient Name	Name of patients that visit the hospital
Age	Age in years at the time of patient's visit
Gender	Patients' gender as recorded
Admission Date	Dates the patient was admitted
Discharge Date	Dates the patient was discharged
Department	Departments where patients were treated
Diagnosis	Patients' diagnoses recorded
Total Bill Amount ($)	Total amount charged for treatment and services for each patient's visit
Insurance Coverage (%)	Share of the bill covered by the insurance
Payment Status	Paid, Pending or Partially Paid

Columns I added during the course of the analysis:

Terms	**Meaning**
Length of Stay	*Formula =DAYS(Discharge Date, Admission Date)* This is the number of days a patient stayed per visit
Age Group	Children(1-17), Young Adult(18-30), Adult(31-60) Elderly(above 60), based on Age

Insurance Amount	Insurance Coverage % x Total Bill Amount. This is the part of the bill that insurance pays per visit
Bill patient is paying	Total Bill Amount – Insurance Amount This is the part the patient owes per visit

Problem Statement

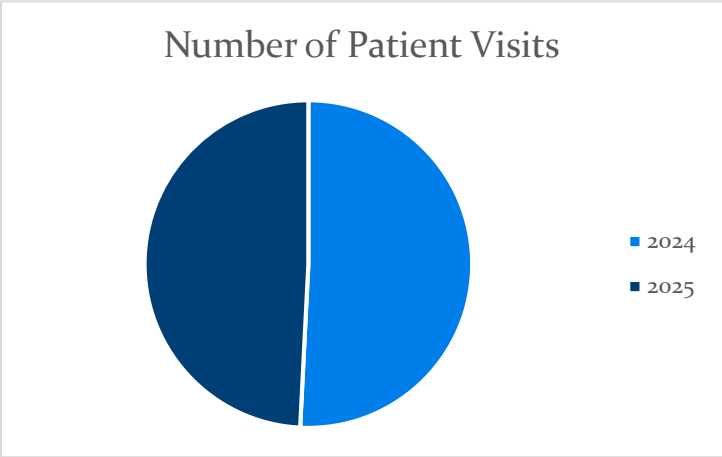
The management of a Secure Life General Hospital has observed a change in patient admissions, outstanding medical bills, and variations in insurance coverage across its departments. The hospital currently lacks a consolidated report that provides a clear picture of its operational and financial performance. As the data analyst, my task was to analyze the data and answer ten management questions:

- How many patient visits has the hospital recorded?
- Which departments receive the highest number of patients?
- Which departments generate the highest total bills?
- What is the average length of stay for patients across different departments?
- What proportion of patient bills has been paid, remains pending, or has been partially paid?
- How much has the hospital billed, and how much remains outstanding?
- How does insurance coverage vary across departments and diagnoses?
- Which age groups account for the highest number of patient visits?
- How do admissions and billing amounts change over time?
- Which areas may require management attention based on the analysis?

Insights

1. Patient Visits

The hospital recorded 1,000 patient visits: 508 in 2024 and 492 in 2025.



2. Patients Volume per department

General department received the highest number of patients with 25.20% while Neurology department received the lowest number of patients with 9.40%.

Department	Visits	Share
General	252	25.2%
Emergency	190	19.0%
Orthopedics	159	15.9%
Pediatrics	153	15.3%
Cardiology	152	15.2%
Neurology	94	9.4%

3. Total bills per department

The Orthopedics and Cardiology departments generated the highest total bills as shown in the table below. Orthopedics and Cardiology together account for 57.7% of all billing but only 31.1% of visits, because their cases are far more expensive. Adding Neurology, whose average bill ($12,929) is close to Cardiology's, the three departments account for 72.8% of billing from 40.5% of visits.

Department	Total billed	Share of billing	Average bill
Orthopedics	$2,553,862	31.7%	$16,062
Cardiology	$2,095,438	26.0%	$13,786
Neurology	$1,215,303	15.1%	$12,929
General	$993,027	12.3%	$3,941
Emergency	$622,525	7.7%	$3,276
Pediatrics	$572,988	7.1%	$3,745

4. Average length of stay

The overall average length of stay for patients is 4.8 days and it ranges from 2.0 (Emergency) to 8.4 (Orthopedics). Stays vary widely by department:

Department	Average stay (days)

Orthopedics	8.4
Neurology	6.9
Cardiology	6.5
General	3.9
Pediatrics	3.0
Emergency	2.0

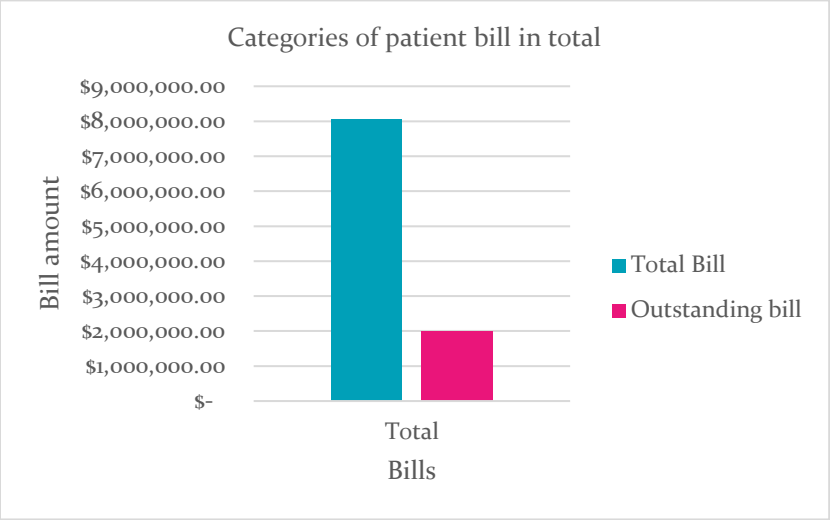
5. Payment Status

56.4% of bills are Paid, 25.7% are Pending and 17.9% are Partially Paid. Paid bills total $4,568,527, Partially Paid $1,401,830 and Pending $2,082,786.

6. Amount billed and outstanding

The hospital billed a total of $8,053,143.44. Insurance covers $6,062,666 of this (75.3%) and patients are responsible for $1,990,477.

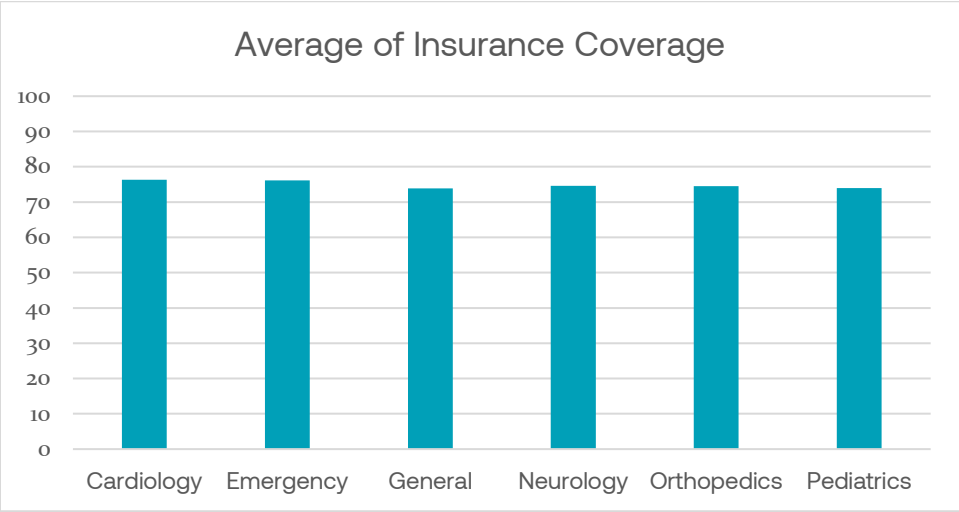
Outstanding here means the patient's share of bills that are still Pending: $580,265.04. This is a deliberate assumption: the data does not say whether insurers have paid their share of Pending bills, so the full $2,082,786 value of Pending bills could also be viewed as outstanding. The data also does not say how much of each Partially Paid bill has been settled. The patient share of those bills is $447,157, so the amount at risk could be as high as $1,027,422.

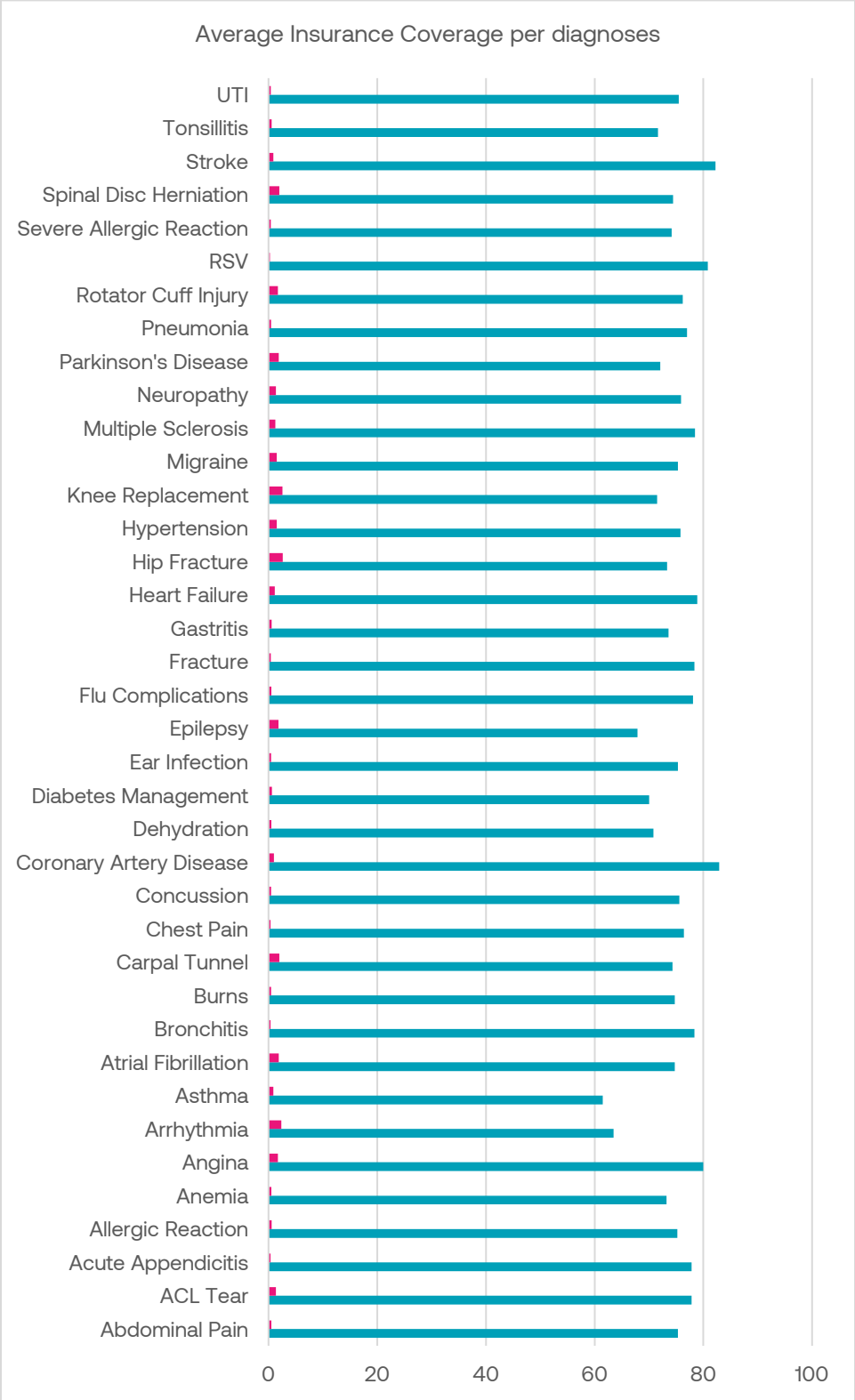


7. Insurance Coverage

Average coverage is similar across departments, between 73.9% (General) and 76.3% (Cardiology). The real differences are between diagnoses: Asthma has the lowest average coverage (61.5%) followed by Arrhythmia (63.5%) and Epilepsy (67.9%), while Coronary Artery Disease (82.9%), Stroke (82.2%) and RSV (80.8%) have the highest. In total, 44 visits had no insurance coverage at all.

Because Orthopedics bills are so large, patients there owe the most: $648,030, against $478,933 in Cardiology.

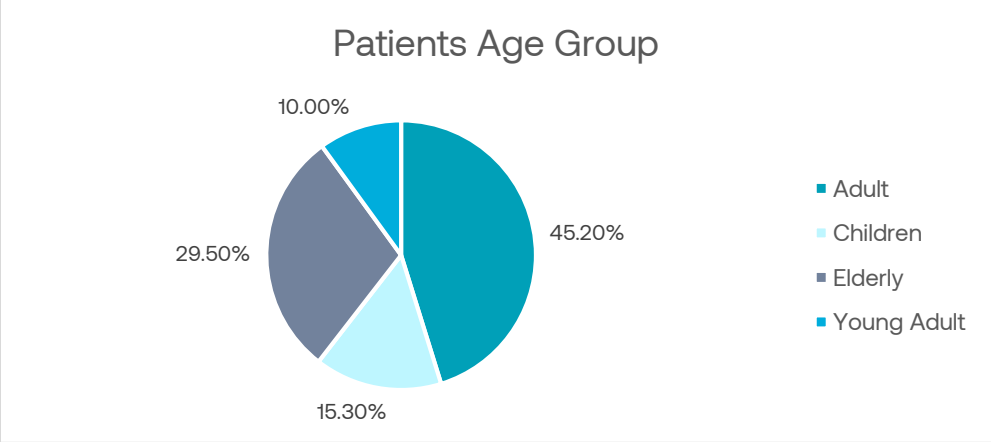



Average Insurance Coverage per diagnoses
UTI
Tonsillitis
Stroke
Spinal Disc Herniation
Severe Allergic Reaction
RSV
Rotator Cuff Injury
Pneumonia
Parkinson's Disease
Neuropathy
Multiple Sclerosis
Migraine
Knee Replacement
Hypertension
Hip Fracture
Heart Failure
Gastritis
Fracture
Flu Complications
Epilepsy
Ear Infection
Diabetes Management
Dehydration
Coronary Artery Disease
Concussion
Chest Pain
Carpal Tunnel
Burns
Bronchitis
Atrial Fibrillation
Asthma
Arrhythmia
Angina
Anemia
Allergic Reaction
Acute Appendicitis
ACL Tear
Abdominal Pain
0
20
40
60
80
100

8. Visits by Age Group

The ages of the patients were grouped into the following: Children (1–17), Young Adult (18–30), Adult (31–60) and Elderly (above 60).

Age group	Visits	Share
Adult (31–60)	452	45.2%
Elderly (60+)	295	29.5%
Children (1–17)	153	15.3%
Young Adult (18–30)	100	10.0%



9. Admissions & Billing amount changes over time

Visits fell slightly from 508 in 2024 to 492 in 2025, and billing fell 5.1% from $4,132,465 to $3,920,678.

Quarterly billing was highest in Q4 of both years (about $1.13M in 2024 and $1.12M in 2025) and lowest in Q1 2024 ($843,958). By month, November was the busiest (100 visits) and February the quietest (66). These trends can be explored using the timeline attached to the

Number of Patient Visits chart and the Categories of Patient Bill chart on the dashboard.

10. Areas needing attention

- **High-cost departments:** Orthopedics, Cardiology and Neurology make up 72.8% of billing from only 40.5% of visits. Their average bills are $16,062, $13,786 and $12,929, against about $3,300 to $3,900 in General, Emergency and Pediatrics. They also have the longest stays, at 8.4 (Orthopedics), 6.9 (Neurology) and 6.5 (Cardiology) days.
- **Collection gaps:** Pediatrics has the highest share of pending bills (28.8%) and General is close behind (27.0%). Emergency collects best, with 62.1% of bills paid.
- **Unpaid bills:** Patients owe about $580k on Pending bills and $447k on Partially Paid bills. Together that is more than $1M of patient-owed money not yet collected, so I recommend following it up.
- **Low coverage:** Asthma (61.5%), Arrhythmia (63.5%) and Epilepsy (67.9%) patients face the largest share of their own bills. Patients with lower coverage may find it harder to pay, so these are good candidates for payment plans.

11. Additional insights

- **Gender:** Visits are almost evenly split, 512 female (51.2%) and 488 male (48.8%). Average bills ($7,996 female, $8,113 male) and average stays (4.8 days for both) are nearly identical, so gender is not a driver of cost or workload.
- **Age and cost:** Adults have the highest average bill ($9,433), followed by the Elderly ($8,831), Young Adults ($6,112) and Children ($3,745).

Recommendation

- Follow up on Pending bills first, starting with Pediatrics and General where the rate is highest, and set up regular payment reminders.
- Track Partially Paid bills with an actual balance remaining, so the true outstanding amount can be reported.

- Review pricing and length of stay in Orthopedics and Cardiology (58% of billing) and Neurology (73% combined) for efficiency savings.
- Help patients with low-coverage diagnoses (such as Asthma and Arrhythmia) and uninsured patients with payment plans and insurance guidance.
- Plan staffing and bed capacity for the Q4 peak and for November, the busiest month.

Conclusion

Secure Life General Hospital handled 1,000 visits and billed $8.05M over two years. Billing is concentrated in Orthopedics, Cardiology and Neurology, while General and Emergency handle the most patients at much lower cost. Insurance covers about 75% of charges, yet roughly $0.6M to $1.0M in patient-owed bills remains unpaid. Focusing on collections, high-cost departments and low-coverage diagnoses would give management the biggest improvement in financial performance. Future work could add a unique patient identifier to measure repeat visits and track partial payment balances.